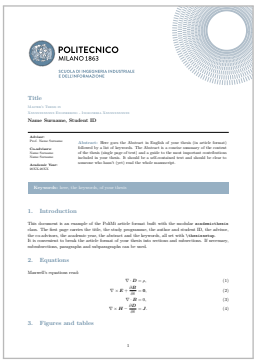
academicthesis

Politecnico di Milano

User and developer guide



`classical`



`article`



`summary`



`phd`

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academicthesis v0.2.0 • October 2026

Contents

I User guide **3**

1 What this package does **4**
1.1 The four formats . . . 4
1.2 What you get . . . 5
1.3 How it is organised . . . 5

2 Installing and compiling **6**
2.1 Requirements . . . 6
2.2 The files you need . . . 6
2.3 Three ways to set it up . . . 6
2.4 Compiling . . . 7

3 Writing your thesis **8**
3.1 The shape of a thesis file . . . 8
3.2 Class options . . . 9
3.3 The data of your thesis . . . 9
3.4 PoliMi options: `\polimithesissetup` . . . 10

4 The formats in detail **11**
4.1 classical . . . 11
4.2 article . . . 11
4.3 summary (Executive Summary) . . . 11
4.4 phd (doctoral dissertation) . . . 12
4.5 Colours, theorems, extra packages . . . 12

5 Front matter and bibliography **13**
5.1 Abstract, originality, acknowledgements, dedication . . . 13
5.2 Bibliography . . . 13
5.3 How the blocks look at PoliMi . . . 14

6 Complete examples **15**
6.1 classical . . . 15
6.2 article . . . 15
6.3 summary . . . 16
6.4 phd . . . 16

7 Troubleshooting **18**

II Developer guide **19**

8 How the core works **20**
8.1 Design principles . . . 20
8.2 Files . . . 21
8.3 What happens when a thesis is compiled . . . 22

8.4 The early configuration file . . . 22
8.5 Thesis data . . . 23
8.6 Registries . . . 24
8.7 Strings and languages . . . 24
8.8 Front matter, second page, page styles . . . 25
8.9 Two rules for writing layout code . . . 25
8.10 Search path in the repository . . . 25
8.11 Messages . . . 26
8.12 Adding a university, in short . . . 26

9 The PoliMi package **27**
9.1 Files . . . 27
9.2 The early configuration . . . 27
9.3 The package . . . 28
9.4 The layout files . . . 28
9.5 Strings . . . 29
9.6 Common changes . . . 29

Licence and credits **30**

Part I

User guide

For students writing a thesis, an Executive Summary or a doctoral dissertation at Politecnico di Milano: what the class does, how to set it up, every option and command available, and complete examples.

Chapter 1

What this package does

Politecnico di Milano provides LaTeX templates for its theses. Using a template means copying a folder, editing its configuration by hand and keeping track of which file prints what. academicthesis turns those layouts into a class: you choose the format with one option, give the data of the thesis once, and the class produces the cover, the headings and the page layout of that format.

```
\documentclass[university=polimi, master, format=classical]{academicthesis}
```

The PoliMi layouts are Politecnico di Milano's design

The layouts produced by this package are **not original work of academicthesis**. They reproduce the official LaTeX templates provided by Politecnico di Milano:

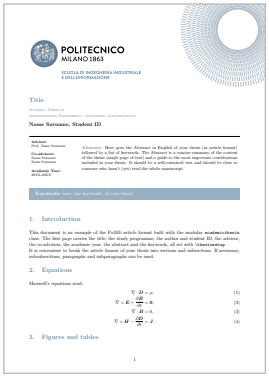
- the thesis templates of the School of Industrial and Information Engineering (Scuola di Ingegneria Industriale e dell'Informazione, "Scuola 3I"), revision of October 2021: *Thesis classical format*, *Thesis article format* and *Executive Summary*, the models of the `classical`, `article` and `summary` formats;
- the PoliMi doctoral dissertation template (`phdthesis.sty`, `phdtitle.sty`, by J. Stevens and L. Fossati), the model of the `phd` format.

The page designs, the colour `bluepoli`, the cover arrangements, the heading styles, the "raggiera" background and the Politecnico logos and seal belong to Politecnico di Milano and to the authors of those templates. This package re-implements them on the shared academicthesis core, so that they are selected with a class option and share its data handling, checks and multilingual blocks.

1.1 The four formats



`classical`



`article`



`summary`



`phd`

Format	What it is	Degrees	Built on
`classical`	The full thesis as a book: cover page, abstracts, chapters. Default for bachelor and master.	bachelor, master	book, 12pt, two-sided, A4
`article`	The thesis as a paper: a header block with the abstract on page 1, then sections.	bachelor, master	article, 11pt, A4
`summary`	The Executive Summary, in two columns under a blue page header.	bachelor, master	article, 11pt, 2 columns, A4
`phd`	The doctoral dissertation. Default for the `phd` degree.	PhD	book, 10pt, 17×24 cm

1.2 What you get

- **The right base class, page size and layout** for the chosen format, chosen for you.
- **The cover or first-page block** of the format, filled from your data.
- **PoliMi headings, colours and captions**: blue chapter and section titles, blue caption labels, theorem and proposition environments in the PoliMi style.
- **Labels in PoliMi's wording** ("Advisor", "Student ID", "Master's Thesis in") in English or Italian, following the language of the thesis.
- **An English and an Italian abstract**, with the correct headings and hyphenation, without extra options.
- **Checks**: if a field the format needs is missing, compilation stops with a message naming it.

1.3 How it is organised

The class has a core shared by every university and one package per university. With `university=polimi` the core loads the PoliMi package by itself, and the PoliMi package loads the code of the chosen format. Part II of this guide explains the code.

Chapter 2

Installing and compiling

2.1 Requirements

- A recent TeX distribution: TeX Live or MacTeX 2023 or later, or an up-to-date MiKTeX. The class is written in LaTeX3 and needs a modern LaTeX kernel.
- **pdfLaTeX** is the reference engine, used by all the examples in this guide. XeLaTeX and LuaLaTeX are also accepted (the class then loads fontspec).
- For the bibliography, biblatex and biber, both part of every full TeX distribution. Without them the class falls back to natbib and BibTeX (Section 5.2).

2.2 The files you need

academicthesis is a collection of small packages that load one another *by name*. For a PoliMi thesis you need only these:

Files	Role
`core/academicthesis.cls`, `core/academicthesis-kernel.sty`	The core class, shared by every university.
`languages/academicthesis-<language>.ldf`	The words printed on the pages, one file per language (English, Italian, French).
`universities/polimi/*`	The PoliMi package: its layout, labels and logos.

The other folders of `universities/` belong to other universities and can be left out.

2.3 Three ways to set it up

1. **Inside the repository.** Copy the PoliMi showcase in `showcases/` and keep its first line,

   ```
   \input{academicthesis-repo-paths}
   ```

   which tells LaTeX where the package folders are. Then compile with any editor, or from a terminal with `latexmk`.
2. **Next to your thesis.** Copy the files listed above into the folder of your thesis (without sub-folders) and delete the `\input{academicthesis-repo-paths}` line. This is also how to use the class on Overleaf: upload the files to the project.
3. **Installed in your personal TeX tree**, for example `~/Library/texmf/tex/latex/academicthesis/` on macOS or `~/texmf/tex/latex/academicthesis/` on Linux. Every document on the computer can then use the class, with no extra line.

Editors

Texifier/Texpad, TeXShop, TeXstudio, VS Code (LaTeX Workshop) and Overleaf all work. In Texifier/Texpad, set the typesetter to *TeX Live (pdfLaTeX)*: its built-in engine is too old for this class.

2.4 Compiling

Run pdfLaTeX twice (for the table of contents and cross-references), with biber between the two runs when the thesis has a bibliography. `latexmk` does all of this by itself:

```
latexmk -pdf my-thesis.tex
```

Chapter 3

Writing your thesis

3.1 The shape of a thesis file

```
\documentclass[university=polimi, master, format=classical,
               language=english]{academicthesis}

\usepackage{amssymb, bm}          % your own packages, if any
\usepackage{hyperref}

\thesissetup{                     % the data printed on the cover
  title         = {Title},
  candidate     = {Name Surname},
  idnumber      = {000000},
  degree        = {Mathematical Engineering - Ingegneria Matematica},
  supervisor    = {Prof. Name Surname},
  cosupervisors = {Name Surname, Name Surname},
  academicyear  = {2025-26},
}
\addbibresource{references.bib}

\begin{document}
\frontpage                        % the cover (classical, phd) or page-1 block

\frontmatter
\begin{abstractthesis}[keywords = {first, second}]
  English abstract.
\end{abstractthesis}
\begin{abstractthesis}[language = italian, keywords = {primo, secondo}]
  Abstract in lingua italiana.
\end{abstractthesis}
\tableofcontents

\mainmatter
\chapter{Introduction}
...
\printbibliography
\end{document}
```

3.2 Class options

Option	Values	Meaning
`university`	`polimi`	Selects the PoliMi package. Required.
degree	`bachelor`, `master`, `phd`	Written on its own. Bachelor and master print "Bachelor's / Master's Thesis in" (Italian: "Tesi di Laurea Triennale / Magistrale in"). Default: `bachelor`.
`format`	`classical`, `article`, `summary`, `phd`	The layout. Default: `classical`, or `phd` with the `phd` degree. `phd` requires that degree.
`language`	`english`, `italian`	Language of every label, and hyphenation. Default: `english`.
`other-languages`	e.g. `{french}`	A third language used by single blocks.
`secondpage`		Adds the core's examination and copyright page, which PoliMi does not use (off by default).
`biblatex-options`	e.g. `{style=numeric-comp}`	Options for biblatex (Section 5.2).
anything else	`11pt`, `oneside`, ...	Passed to the base class; your choice wins over the format's default.

3.3 The data of your thesis

All the data is given with `\thesissetup` in the preamble. Write academic titles ("Prof.") yourself.

3.3.1 Bachelor's and master's (classical, article, summary)

Key	Printed as (English / Italian)	classical	article	summary
`title`	Title, in blue	**required**	**required**	**required**
`subtitle`	Under the title	optional	optional	optional
`candidate`	Author / Candidato	**required**	**required**	**required**
`idnumber`	Student ID / Matricola	**required**	**required**	—
`degree`	After "Master's Thesis in"	**required**	**required**	**required**
`supervisor(s)`	Advisor(s) / Relatore, Relatori	**required**	**required**	**required**
`cosupervisor(s)`	Co-advisor(s) / Correlatore, Correlatori	optional	optional	optional
`academicyear`	Academic Year / Anno Accademico	**required**	**required**	**required**
`abstract`	English abstract on page 1	—	**required**	—
`keywords`	Its keywords, in a blue box	—	optional	—
`gender`	Candidato / Candidata (Italian)	optional	optional	optional

3.3.2 Doctoral dissertation (phd)

Key	Printed as (English / Italian)	Required
`title`, `subtitle`	Title; subtitle when given	title
`candidate`	Under “Doctoral Dissertation of” / “Tesi di Dottorato di”	**required**
`department`	In full, as printed: “Department of Electronics, Information and Bioengineering”	**required**
`phdtitle`	In full, as printed: “Doctoral Program in Information Technology”	**required**
`supervisor(s)`	Supervisor(s) / Relatore, Relatori	**required**
`tutor(s)`	Tutor(s) / Tutor	optional
`coordinator`	The Chair of the Doctoral Program / Il Coordinatore del Dottorato	optional
`academicyear`, `cycle`	Bottom line: “2025–2026 – Cycle XXXVIII”	**required**

One person or several

`supervisor` takes one name and keeps any comma in it; `supervisors` takes several, separated by commas, and the label becomes “Advisors”. The same holds for `cosupervisor(s)` and `tutor(s)`. `\thesissetup` can be used several times; a later value replaces an earlier one.

3.4 PoliMi options: `\polimithesissetup`

Key	Default	Effect
`logo-file`	`logopolimi3i`	Logo on the cover or first page, file name without extension. The `phd` format uses the round seal, `logopolimiseal`.
`watermark`	`true`	The “raggiera” in the background of the cover, of page 1 and of blank pages; `false` removes it. Not used by `phd`.
`paper`	`17x24`	`phd` only: `17x24` (as in the template) or `a4`.

```
\polimithesissetup{watermark = false}
```

Chapter 4

The formats in detail

4.1 classical

- **Page**: A4, 12pt, two-sided; inner margin 3 cm, outer 2 cm, top and bottom 2.5 cm; 1.5 line spacing; no paragraph indent.
- **Cover**: Scuola 3I logo, title in blue, "Master's Thesis in *degree*", "Author: *name*", and at the bottom student ID, advisor(s), co-advisor(s) and academic year; raggiera at the top right.
- **Chapters**: a large blue number and bar, "**1**| Chapter title". Starred chapters (`\chapter*{Introduction}`) still enter the table of contents and the running header.
- **Running header**: page number outside, chapter inside.
- **Blank pages** before a chapter carry the raggiera; `\blankpage` inserts one by hand.

Typical order: `\frontpage`, `\frontmatter`, English and Italian abstracts, `\tableofcontents`, `\mainmatter`, chapters, bibliography, `\appendix`, `\backmatter`, lists of figures and tables, acknowledgements.

4.2 article

- **Page**: A4, 11pt, 2 cm margins, no indent; page number centred at the bottom.
- **Page 1**: no separate cover. `\frontpage` prints a header block (logo, title, programme, "author, student ID", a rule, advisors and academic year beside the **abstract**, and the keywords in a blue box); the text starts right after it.
- **Sections** instead of chapters. A `\chapter` still compiles, as a section, with a warning.

The English abstract is therefore data (`abstract`, `keywords` in `\thesissetup`); an Italian abstract goes at the end with `abstractthesis`.

4.3 summary (Executive Summary)

- **Page**: A4, 11pt, two columns 30 pt apart, 2 cm side margins.
- **Page 1**: a full-width block (logo, "Executive Summary of the Thesis", title, programme, author, advisor(s), co-advisor(s), academic year, a rule), then the two columns.
- **Following pages**: a blue box at the top with "Executive summary" and the author; page number centred.

The student ID is not printed and not required. As in the template, the summary usually ends with a short bibliography.

4.4 phd (doctoral dissertation)

- **Page**: 10pt, two-sided, chapters on right-hand pages; 17×24 cm with 2 cm margins, or A4 with `paper=a4`. Times for the text, Helvetica for headings.
- **Cover**: PoliMi seal, "Politecnico di Milano", department and programme in small capitals, a rule, the title, "Doctoral Dissertation of" and the author on the right, supervisor, tutor and chair of the programme on the left, academic year and cycle at the bottom.
- **Chapters** in the fncychap "Sonny" style; sections in bold sans serif with a rule; running headers with chapter (even pages) and section (odd pages).
- A `defn` (Definition) environment numbered by chapter; grey drop caps with `\lettrine`.

Not taken over from the template, on purpose: psboxit (not in TEX Live), fancybox (unused) and its glossaries set-up (its `\glossarystyle` command no longer exists). Load glossaries, crop, listings and so on yourself if you need them.

4.5 Colours, theorems, extra packages

- `bluepoli` (CMYK 0.4, 0.1, 0, 0.4) is defined in every format, and also as `bluePoli` in `article` and `summary`, the name used by the templates: `\rowcolor{bluepoli!40}` with colortbl.
- `theorem` and `proposition` have blue headings, numbered by chapter (`classical`) or section (`article`, `summary`); your own `\newtheorem`s take the same style.
- Left to you, as in the templates: hyperref, cleveref, amssymb, bm, float, colortbl, subfig, tikz, algorithm, enumitem.

Chapter 5

Front matter and bibliography

5.1 Abstract, originality, acknowledgements, dedication

The core class provides four front-matter blocks. They are written the same way at every university; the PoliMi package only sets their look.

Block	Options	Default heading (English)
`\begin{abstractthesis}`	`language`, `keywords`, `title`	Abstract
`\begin{originality}`	`language`, `title`	Statement of Originality
`\begin{acknowledgements}`	`language`, `title`	Acknowledgments
`\dedication[...]{text}`	`language`	none: the text alone on its page

- `language=` writes the block in another language: its heading, the "Keywords" label and the hyphenation follow that language. Without it, the block uses the main language of the thesis. English and Italian are always loaded for PoliMi, so an Italian abstract needs nothing more; for a third language, list it in the `other-languages` class option.
- `keywords=` prints a list of keywords after the abstract.
- `title=` replaces the default heading.

Abstracts, statements of originality and acknowledgements are listed in the table of contents. An English abstract followed by an Italian one:

```
\begin{abstractthesis}[keywords = {corporate finance, valuation}]
  The English abstract.
\end{abstractthesis}
\begin{abstractthesis}[language = italian,
                       keywords = {finanza aziendale, valutazione}]
  L'abstract in italiano.
\end{abstractthesis}
\dedication{To my family.}
```

5.2 Bibliography

The class loads biblatex with biber for you; do not load biblatex again in your preamble. Its style is, in order of priority: the options you give with the class option `biblatex-options`; the university's default (PoliMi sets none, so the numeric style applies); otherwise a numeric style.

```
\documentclass[..., biblatex-options={style=apa}]{academicthesis}
\addbibresource{references.bib}   % in the preamble
...
\printbibliography                % where the bibliography goes
```

If biblatex is not installed, the class warns and uses natbib with BibTeX behind the same two

commands, so the thesis still compiles.

5.3 How the blocks look at PoliMi

Block	classical, phd	article, summary
`abstractthesis`	Unnumbered chapter; keywords as a bold line	Unnumbered section; keywords in a blue box
`originality`	Unnumbered chapter	Unnumbered section
`acknowledgements`	Unnumbered chapter, upright text	Unnumbered section, upright text
`\dedication`	Own page, right-aligned italic	same

With `language=italian`, the abstract heading becomes “Abstract in lingua italiana” and the keywords label “Parole chiave”.

Chapter 6

Complete examples

6.1 classical

```
\documentclass[university=polimi, master, format=classical,
               language=english]{academicthesis}
\usepackage{amssymb, bm, float, colortbl}
\usepackage[colorlinks=true, linkcolor=black, citecolor=black]{hyperref}

\thesissetup{
  title         = {Title},
  candidate     = {Name Surname},
  idnumber      = {000000},
  degree        = {Mathematical Engineering - Ingegneria Matematica},
  supervisor    = {Prof. Name Surname},
  cosupervisors = {Name Surname, Name Surname},
  academicyear  = {2025-26},
}
\addbibresource{references.bib}

\begin{document}
\frontpage
\frontmatter
\begin{abstractthesis}[keywords = {first, second, third}]
  English abstract.
\end{abstractthesis}
\begin{abstractthesis}[language = italian, keywords = {primo, secondo}]
  Abstract in lingua italiana.
\end{abstractthesis}
\tableofcontents

\mainmatter
\chapter*{Introduction}
\chapter{Chapter one}
...
\printbibliography
\appendix
\chapter{Appendix A}

\backmatter
\listoffigures
\listoftables
\begin{acknowledgements} Thanks to ... \end{acknowledgements}
\end{document}
```

6.2 article

```
\documentclass[university=polimi, master, format=article,
```

```
                language=english]{academicthesis}
\thesissetup{
  title        = {Title},
  candidate    = {Name Surname},
  idnumber     = {000000},
  degree       = {Mathematical Engineering - Ingegneria Matematica},
  supervisor   = {Prof. Name Surname},
  academicyear = {2025-26},
  abstract     = {The English abstract, printed on page 1.},
  keywords     = {here, the keywords},
}
\begin{document}
\frontpage
\section{Introduction}
...
\appendix
\section{Appendix A}
\clearpage
\begin{abstractthesis}[language = italian, keywords = {qui, le parole chiave}]
  Abstract in lingua italiana.
\end{abstractthesis}
\begin{acknowledgements} ... \end{acknowledgements}
\end{document}
```

6.3 summary

```
\documentclass[university=polimi, master, format=summary,
                language=english]{academicthesis}
\thesissetup{
  title        = {Title of the thesis},
  candidate    = {Name Surname},
  degree       = {Mathematical Engineering - Ingegneria Matematica},
  supervisor   = {Prof. Name Surname},
  cosupervisor = {Name Surname},
  academicyear = {2025-26},
}
\begin{document}
\frontpage
\section{Introduction}
...
\printbibliography
\end{document}
```

6.4 phd

```
\documentclass[university=polimi, phd, language=english]{academicthesis}
\polimithesissetup{paper = 17x24}            % or a4
\thesissetup{
  title        = {Title},
  candidate    = {Name Surname},
  department   = {Department of Electronics, Information and Bioengineering},
  phdtitle     = {Doctoral Program in Information Technology},
  supervisor   = {Prof. Name Surname},
```

```
  tutor        = {Prof. Name Surname},
  coordinator  = {Prof. Name Surname},
  academicyear = {2025--2026},
  cycle        = {XXXVIII},
}
\begin{document}
\frontpage
\frontmatter
\begin{abstractthesis} The abstract. \end{abstractthesis}
\tableofcontents
\mainmatter
\chapter{Introduction}
...
\end{document}
```

Complete, compilable versions: `showcases/example-polimi-classical.tex`, `-article.tex`, `-summary.tex` and `-phd.tex`.

Chapter 7

Troubleshooting

Message (abridged)	Cause	Fix
Missing required thesis metadata: '*key*'	A required field is not in `\thesissetup`	Add it; the message shows an example. See Section 3.3.
Missing required thesis metadata: 'abstract'	`format=article` without the abstract	Add `abstract` (and `keywords`).
Missing required thesis metadata: 'idnumber'	No student ID (classical, article)	Add `idnumber`.
Missing required thesis metadata: 'department' (or 'phdtitle', 'cycle')	PhD cover data missing	Add it to `\thesissetup`.
Unknown format '*x*' for university 'polimi'	Typo in `format`	Use `classical`, `article`, `summary` or `phd`.
format=phd is the PoliMi doctoral dissertation: use the 'phd' degree	`format=phd` with `bachelor` or `master`	Use the `phd` degree.
No front page registered for university 'polimi', degree 'phd'	`phd` degree with another format	Use `format=phd`, or leave `format` out.
File 'academicthesis.cls' not found	LaTeX cannot see the class files	Keep the line that inputs `academicthesis-repo-paths`, or copy the files next to the thesis (Chapter 2).
Warning: The base class 'article' has no chapters	`\chapter` in `article` or `summary`	Use `\section`.
Warning: 'french' ... was not loaded by babel	A block with `language=french`	Add `french` to the `other-languages` option.

Quick reference

```
\documentclass[university=polimi, bachelor|master,
               format=classical|article|summary,
               language=english|italian]{academicthesis}
\documentclass[university=polimi, phd]{academicthesis}     % format=phd
\polimithesissetup{logo-file=..., watermark=true|false,
                   paper=17x24|a4}                          % paper: phd only
% classical: title candidate idnumber degree supervisor(s) academicyear
% article:   the same + abstract (keywords optional)
% summary:   as classical, without idnumber
% phd:       title candidate department phdtitle supervisor(s)
%            academicyear cycle  (tutor(s), coordinator optional)
\frontpage
\begin{abstractthesis}[language=..., keywords=..., title=...]
  ... \end{abstractthesis}
\begin{acknowledgements}[language=..., title=...]
  ... \end{acknowledgements}
```

Part II

Developer guide

For those who maintain or extend the code: first how the shared core works, then how the PoliMi package and its four formats are built on it.

Chapter 8

How the core works

This chapter describes the part of the code that every university shares. The next chapter builds on it to explain the PoliMi package. The reader is assumed to know LaTeX 2_ε package writing; LaTeX3 (expl3) names are explained where they matter.

8.1 Design principles

The core knows no university. `academicthesis.cls` never mentions LUISS, Politecnico di Milano or Bocconi. A university package plugs into a small number of *registries* (front pages, required fields, bibliography defaults) and *declarations* (base class, formats, second page, extra languages, extra class options) that the core consults at the right moment.

A university knows only the core. A university package uses the kernel's public commands and never another university's code, so each one can be versioned, shipped and installed on its own.

Files are found by name, never by folder. Every `\LoadClass`, `\RequirePackage`, `\input` and `\includegraphics` names a file, not a path. Once installed in a TeX tree, kpathsea finds each file whatever package shipped it; in the repository, the search path is extended instead of changing the code (Section 8.10).

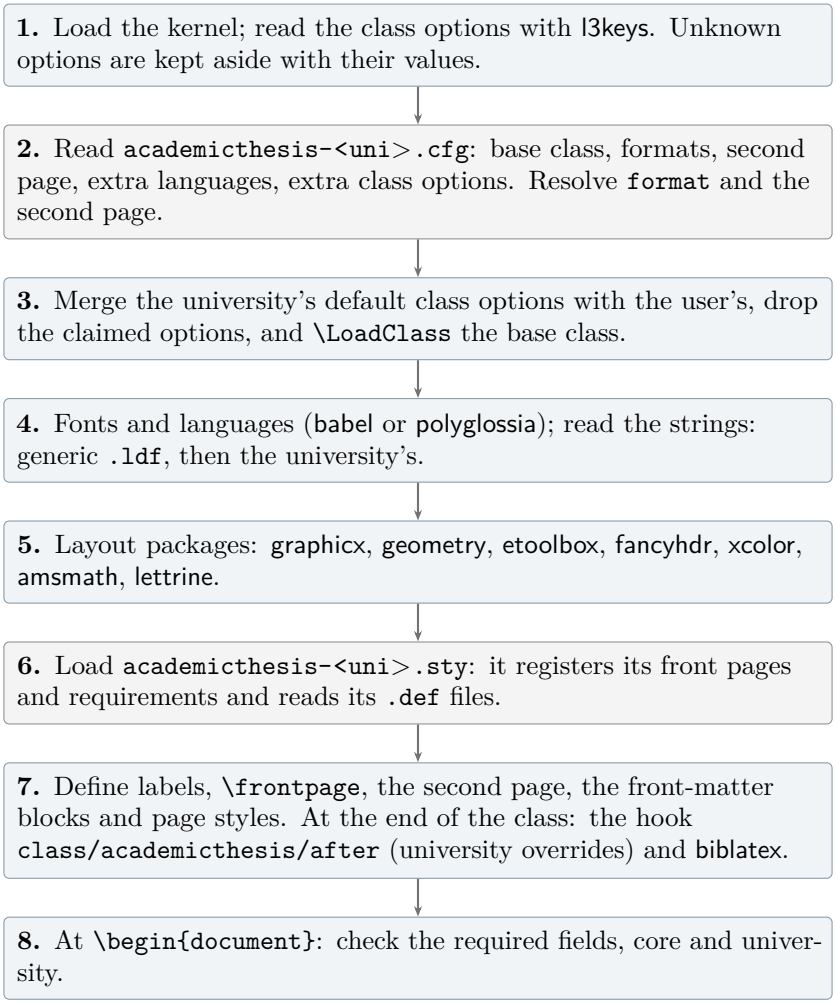
One data store, one vocabulary. All thesis data lives in a single property list with the same key names at every university. Universities change *wording* through string files, not key names.

Logic in LaTeX3, typesetting in LaTeX 2_ε. Option handling, registries and validation are written in expl3; everything that puts text on the page is written in ordinary LaTeX syntax, where spaces and blank lines behave as usual (Section 8.9).

8.2 Files

File	Role
`core/academicthesis.cls`	The class: options, base class, fonts and languages, bibliography, loading the university, labels, `\frontpage`, second page, front-matter blocks, page styles, validation.
`core/academicthesis-kernel.sty`	The shared API: messages, metadata store, `\thesissetup`, lists, registries, early-configuration commands, university loader, string store. Loaded first, by the class.
`languages/academicthesis-<lang>.ldf`	Generic strings for one language (`\ProvideThesisString` calls only).
`universities/<uni>/academicthesis-<uni>.cfg`	Early configuration, read *before* the base class is loaded.
`universities/<uni>/academicthesis-<uni>.sty`	The university package proper.
`universities/<uni>/academicthesis-<uni>-*.def`	Layout code in normal LaTeX syntax, read by the `.sty`.
`universities/<uni>/academicthesis-<uni>-<lang>.ldf`	Optional: the university's wording of core strings, and its own strings.
`showcases/academicthesis-repo-paths.tex`	Lets the examples find the repository folders (not part of any package).

8.3 What happens when a thesis is compiled



Grey steps are where the university's own files run. Two consequences matter when writing a university package:

- The `.cfg` runs before `\LoadClass`: it may only use the early-configuration commands (Section 8.4) and expl3 logic, not packages or document commands.
- The `.sty` runs *before* the core defines its labels, blocks and page styles (step 7). To change one of those, a university wraps its redefinition in the hook that LaTeX runs once the class has finished:

```
\AddToHook{class/academicthesis/after}{%
  \RenewDocumentCommand\thesiskeywords{mm}{...}}
```

8.4 The early configuration file

`academicthesis-<uni>.cfg` is read with expl3 syntax on (spaces are ignored; write `~` for a space). Its commands concern only the active university, so they take no university argument.

Command	Meaning
`\DeclareThesisBaseClass {<class>}{<options>}`	Base class (book, article, ...) and its default options. A default is dropped when the user chose another option of the same group: `11pt` beats a default `12pt`, `oneside` beats `twoside`. Default: book with no options.
`\DeclareThesisFormats {<list>}{<default>}`	The values accepted by `format=`. Resolved at once, so the rest of the file can branch on `\g__academicthesis_format_tl`. A university that declares none rejects `format=`.
`\DeclareThesisSecondPage {true|false}`	Default for the second page (examination commission and copyright). The user can override it with `secondpage` / `no-secondpage`. Default: `true`.
`\DeclareThesisOtherLanguages {<list>}`	Languages always loaded with babel, besides the main one.
`\DeclareThesisClassOption {<name>}{<default>}`	Claims a `\documentclass` option for the university; it is no longer passed to the base class. `name` alone means `true`. Read it with `\academicthesis_class_option:n{<name>}` or, in normal syntax, `\thesisclassoption{<name>}`.
`\DeclareThesisCoreRequirements {true|false}`	Whether the core checks its own required fields (title, candidate, academic year, degree, supervisors). The university's requirements are checked either way. Default: `true`.

8.5 Thesis data

8.5.1 The store

All data given with `\thesissetup` goes into one global property list. `\thesissetup` is a thin l3keys front-end: each key writes one entry.

expl3 (in .sty/.cfg)	Effect
`\academicthesis_meta_set:nn {key}{value}`	Store a value.
`\academicthesis_meta_get:n {key}`	The value (expandable; empty when missing).
`\academicthesis_meta_if_empty:nTF`	Test for an empty or missing value (also `p`, `T`, `F`).
`\academicthesis_list_set:nn`, `\_set_csv:nn`, `\_add:nn`	Set a list to one item, to comma-separated items, or append one.
`\academicthesis_list_count:n`, `\academicthesis_list_if_empty:nTF`	Number of items; emptiness test.

Normal syntax (in .def)	Effect
\thesismeta {key}	The value (expandable).
\IfThesisMetaEmptyTF {key}{empty}{not empty}	Branch on a value.
\academicthesislist [sep]{list}	Typeset a list; default separator \\.
\thesislistcount {list}, \IfThesisListEmptyTF	Count items; branch on emptiness.
\IfThesisDegreeTF {bachelor|master|phd}{yes}{no}	Branch on the degree.
\thesisclassoption {name}	Value of a claimed class option.

The standard keys are title, subtitle, candidate, idnumber, degree, chair, department, major, phdtitle, cycle, academicyear, copyyear, examdate, authoremail, website, dedication, abstract, keywords, coordinator and gender, plus the lists supervisors, cosupervisors, examiners and tutors (each with a singular key that sets one item without splitting at commas). A university may store keys of its own with \academicthesis_meta_set:nn; the core never reads them.

8.5.2 Validation

At \begin{document} the core checks its own fields (unless switched off in the .cfg) and then every entry registered with

```
\DeclareThesisRequirement{<uni>}{<boolean>}{<key>}{<example>}
```

whose university is the active one and whose expl3 boolean is true: \c_true_bool (always), \g__academicthesis_bachelor_bool, _master_bool or _phd_bool. A missing field raises the error *Missing required thesis metadata: '<key>'*, whose help text shows <example>.

8.6 Registries

Command	Meaning
\DeclareThesisFrontPage {<uni>}{<degree>}{<cs>}	\frontpage looks up <uni>.<degree> and runs <cs>; if nothing is registered, it raises *No front page registered*. The command typesets its own page(s).
\DeclareThesisRequirement {<uni>}{<bool>}{<key>}{<ex>}	A required field (above).
\DeclareThesisBibDefaults {<uni>}{<options>}	biblatex options used when the user gives no biblatex-options.

8.7 Strings and languages

Every printed word comes from a string store, one per language, filled by \ProvideThesisString{<key>}{<text>}. For each language the kernel reads academicthesis-<lang>.ldf and then, if it exists, academicthesis-<uni>-<lang>.ldf.

A university file reuses the core keys to change the wording (for instance `SupervisorLabel`) and may add keys of its own.

`\thesisstring{<key>}` prints a string in the language currently selected for strings: the main language, or the one a block switched to with `language=`. It falls back to the main language, then to the key name, so a missing translation never stops compilation.

The labels `\authorlabel`, `\supervisorlabel` and `\cosupervisorlabel` choose the singular or plural string from the length of the list, and the female form of the candidate label from `gender`.

8.8 Front matter, second page, page styles

- `\frontpage` dispatches to the registered front page, then prints the second page if it is on.
- `abstractthesis`, `originality`, `acknowledgements` and `\dedication` all go through one option parser (`language`, `title`, `keywords`). Headings use `\thesisfrontchapter`, which writes `\chapter*` or `\section*` depending on whether the base class has chapters, and adds the table-of-contents entry.
- Universities can restyle without touching the parser: `\thesiskeywords{<label>}{<keywords>}` prints the keywords; `\thesisacknowledgementsbegin{<heading>}` and `\thesisacknowledgementsend` frame the acknowledgements.
- The core defines the page styles `Thesis`, `acknowledgments` and `plain`; `\frontmatter` switches to `plain` and `\mainmatter` to `Thesis`. Under a base class without chapters, the core supplies light versions of `\frontmatter`, `\mainmatter` and `\backmatter`.

8.9 Two rules for writing layout code

Typeset in normal syntax. Under `\ExplSyntaxOn` spaces are ignored and blank lines do not end paragraphs, which silently breaks any text. University packages therefore keep their expl3 code in the `.sty` and put pages in `.def` files read with the document catcodes:

```
\cs_new_protected:Npn \__myuni_input_layout:n #1
  {
    \cctab_begin:N \c_document_cctab   % no group: definitions survive
      \file_input:n { academicthesis-myuni-#1.def }
    \cctab_end:
  }
```

Never load a package from a `.def` file. When a package loaded from there ends, LaTeX restores the syntax of the class that is loading everything, expl3, and every space that follows is lost. The `.sty` loads the packages before reading the `.def`.

8.10 Search path in the repository

Inside the repository the files sit in sub-folders, which TeX does not search. Two mechanisms make them visible without any path in the code:

- **latexmk**: the root `latexmkrc` adds `core//`, `languages//` and `universities//` to `TEXINPUTS` (the trailing `//` also searches sub-folders).

- **Any editor**: `showcases/academicthesis-repo-paths.tex`, input before `\documentclass`, extends LaTeX's `\input@path` with the folders. `\input@path` does not search sub-folders, so it lists each university folder; a new university needs a line there.

8.11 Messages

All messages use the expl3 message module `academicthesis`, so they read *Class academicthesis Error* and carry a help text.

Name	When
`missing-metadata`	A required field is empty.
`missing-frontpage`	No front page registered for the university and degree.
`unknown-university`	No `academicthesis-<uni>.sty` found.
`missing-config`	The `.sty` exists but the `.cfg` does not.
`unknown-format`, `no-formats`	A `format` the university does not declare, or any `format` at a university with none.
`missing-language`	No `.ldf` for the chosen language (falls back to English).
`bibliography-fallback`	biblatex unavailable (warning).

A university defines its own messages in the same module with `\msg_new:nnnn{academicthesis}{<uni>-<name>}{...}{...}`.

8.12 Adding a university, in short

1. Create `universities/<uni>/` with `academicthesis-<uni>.cfg` (at least `\DeclareThesisBaseClass`).
2. Write `academicthesis-<uni>.sty`: `\RequirePackage {academicthesis-kernel}`, setup keys, packages, requirements, then read the `.def` files.
3. In each `.def`, define the front page and register it with `\DeclareThesisFrontPage` for each degree offered.
4. Add `academicthesis-<uni>-<lang>.ldf` files for its wording.
5. Add its folder to `showcases/academicthesis-repo-paths.tex` and a showcase.

Nothing in `core/`, `languages/` or another university changes.

Chapter 9

The PoliMi package

Origin of the layout code

The layout files re-implement Politecnico di Milano's official templates (Scuola 3I, October 2021; PhD template by J. Stevens and L. Fossati). Measurements, colours and heading styles are taken from those templates; when changing them, compare with the originals.

9.1 Files

File	Role
`academicthesis-polimi.cfg`	Formats, base class per format, the `phd` rule, no second page, English and Italian always loaded.
`academicthesis-polimi.sty`	`\polimithesissetup`, requirements per format, packages per format, reading the format's `.def`.
`academicthesis-polimi-classical.def`	Classical layout.
`academicthesis-polimi-article.def`	Article layout.
`academicthesis-polimi-summary.def`	Executive Summary layout.
`academicthesis-polimi-phd.def`	Doctoral dissertation layout.
`academicthesis-polimi-english.ldf`, `-italian.ldf`	PoliMi wording of core strings, and PoliMi-only strings.
`logopolimi3i.pdf`, `raggierapolimi.pdf`, `logopolimiseal.png`	Scuola 3I logo, background, PhD seal (converted from the templates).
`academicthesis-polimi-doc.tex`	The package's own short manual.

9.2 The early configuration

The `.cfg` is where the four formats become four base classes:

```
% the phd degree defaults to the phd format
\bool_if:NT \g__academicthesis_phd_bool
  { \tl_if_empty:NT \g__academicthesis_format_tl
      { \tl_gset:Nn \g__academicthesis_format_tl { phd } } }
\DeclareThesisFormats { classical , article , summary , phd } { classical }
% format=phd needs the phd degree (error polimi-phd-needs-phd)

\str_case:VnF \g__academicthesis_format_tl
  {
    { classical } { \DeclareThesisBaseClass { book }    { 12pt , twoside , a4paper
    } }
```

```
    { article }   { \DeclareThesisBaseClass { article } { 11pt , a4paper } }
    { summary }   { \DeclareThesisBaseClass { article } { 11pt , a4paper ,
    twocolumn } }
    { phd }       { \DeclareThesisBaseClass { book }    { 10pt , twoside ,
    openright } }
  }
  { ... }
\DeclareThesisSecondPage { false }
\DeclareThesisOtherLanguages { english , italian }
```

Because `\DeclareThesisFormats` resolves the format at once, the rest of the file can branch on it. The core then merges these defaults with the user's options and loads the class.

9.3 The package

1. Options. `\polimithesissetup` sets the key family `academicthesis-polimi/layout`: `logo-file` (default chosen by format: `logopolimiseal` for `phd`, `logopolimi3i` otherwise), `watermark` (the switch `\ifpolimi@watermark`) and `paper` (runs `\polimi@paper@17x24` or `\polimi@paper@a4`, defined by the PhD layout). `\polimi@logofile` gives the logo to the layout files.

2. Requirements. On top of the core's (title, candidate, degree, supervisors, academic year): `idnumber` for `classical` and `article`; `department`, `phdtitle` and `cycle` for `phd`; `abstract` for `article`.

3. Packages per format, loaded from the `.sty` (never from a `.def`):

Format	Packages
`classical`	titlesec, caption, setspace, eso-pic, transparent, amsthm
`article`	titlesec, caption, eso-pic, transparent, amsthm, tcolorbox
`summary`	titlesec, caption, eso-pic, transparent, amsthm, tcolorbox
`phd`	times, caption, amsthm, fncychap (Sonny), titlesec (calcwidth), tocloft (titles)

4. The layout. Finally the `.sty` reads `academicthesis-polimi-<format>.def` with the document catcodes.

9.4 The layout files

All four follow the same plan, in numbered sections:

1. **Colour, page, paragraphs**: `bluepoli`, `\geometry`, spacing, drop-cap colour.
2. **Captions and theorems**: the `polimi` theorem style, `theorem` and `proposition` (PhD: `defn`), labels from `\thesisstring`.
3. **Headings** with titlesec. The classical format also redefines `\chapter` so that starred chapters reach the table of contents and the running header.
4. **Watermark**: the raggiera placed with eso-pic (`\AddToShipoutPicture*`) on the cover, page 1 or blank pages; the classical format redefines `\cleardoublepage` for that.
5. **Front page**: `\polimi@frontpage`, registered for the degrees of that format:

```
\DeclareThesisFrontPage{polimi}{bachelor}{\polimi@frontpage}
\DeclareThesisFrontPage{polimi}{master}{\polimi@frontpage}
% academicthesis-polimi-phd.def:
\DeclareThesisFrontPage{polimi}{phd}{\polimi@frontpage}
```

6. **Core pieces redefined**, inside `\AddToHook{class/academicthesis/after}{...}`: the page styles `Thesis` and `plain`, `\thesisfrontchapter`, `\thesiskeywords` (the blue box in `article` and `summary`) and the acknowledgements frame.

Metadata is read with `\thesismeta`, `\IfThesisMetaEmptyTF`, `\IfThesisListEmptyTF` and `\IfThesisDegreeTF`; labels with `\thesisstring`.

9.5 Strings

`academicthesis-polimi-<lang>.ldf` changes the wording of core keys (`CandidateLabel` = Author, `IDNumber` = Student ID, `SupervisorLabel` = Advisor, `CoSupervisorLabel` = Co-advisor) and adds PoliMi keys: `BachelorThesisIn`, `MasterThesisIn`, `CoverAbstractLabel`, `ExecutiveSummaryOfThesis`, `ExecutiveSummaryHeader`, `DoctoralDissertationOf`, `PhDSupervisorLabel(s)`, `TutorLabel(s)`, `CoordinatorLabel`, `DefinitionLabel`, `TheoremLabel`, `PropositionLabel`.

9.6 Common changes

To change…	**Edit…**
a cover or first-page block	section 5 (PhD: 4) of that format's `.def`.
headings, captions, theorems	sections 2–3 of that format's `.def`.
running headers, keywords, acknowledgements	the `class/academicthesis/after` hook of that `.def`.
a label's wording	`academicthesis-polimi-<lang>.ldf`.
required fields	section 2 of the `.sty`.
default class options of a format	the `.cfg`.

Adding a format. Add it to `\DeclareThesisFormats` and to the base-class switch in the `.cfg`, list its packages in the `.sty`, and write `academicthesis-polimi-<format>.def` with its front page registered by `\DeclareThesisFrontPage`.

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The layouts of the PoliMi package are based on the official LaTeX templates of Politecnico di Milano: the Scuola 3I thesis templates (classical format, article format, Executive Summary; revision of October 2021) and the PoliMi doctoral dissertation template by J. Stevens and L. Fossati. Their design, and the Politecnico di Milano name, logos, seal and graphic elements, belong to Politecnico di Milano and to the templates' authors. academicthesis is not published by Politecnico di Milano: before submitting, check the current rules of your school or doctoral programme.